Practical - 13

Practical Title: PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Problem Definition: Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload. Show loading and error states.

A. Theory & Analysis

1. Do PHP endpoints return valid JSON responses?

Yes. The backend script api/events.php sets header 'Content-Type: application/json; charset=utf-8' and outputs responses with json_encode(['success' => true, 'data' => ...]). Proper HTTP status codes (200, 400, 403, 405, 500) are returned based on the operation.

2. Are GET and POST requests handled properly?

Yes. GET requests fetch, search, and filter event records from the database. POST requests handle mutations (create, update, delete) parsed from JSON payloads or form submissions, with admin authorization enforced via session checks.

3. Are loading, success, and error states shown on the frontend?

Yes. The frontend client in pages/events_api.php shows a spinning loading indicator during network requests, disables submit buttons to prevent duplicate clicks, and renders alert banners on success or error without refreshing the browser.

4. Is input validation performed before database operations?

Yes. Request parameters are checked for required fields, valid date formats, and recognized categories before executing parameterized SQL statements.

B. Implementation Details

I developed a REST-style PHP JSON API in api/events.php supporting GET (search/filter/sort) and POST (create/update/delete) with audit logging. I built a single-page interface in pages/events_api.php that consumes this API asynchronously using JavaScript fetch(), performing real-time CRUD operations without page reload, confirmed via browser Network inspection.

C. Evidence of Work

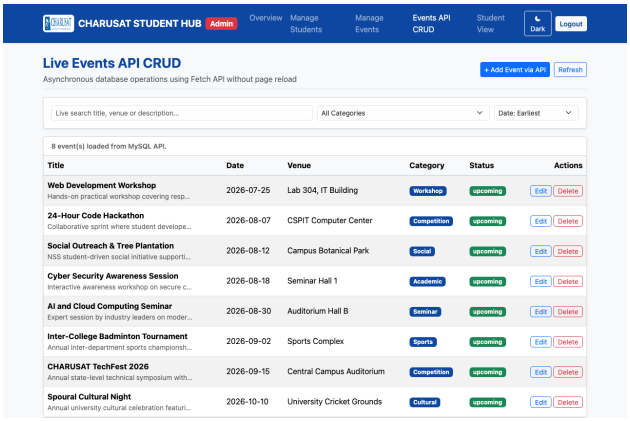


Figure 1: Asynchronous Event CRUD Interface Consuming PHP JSON API via Fetch API

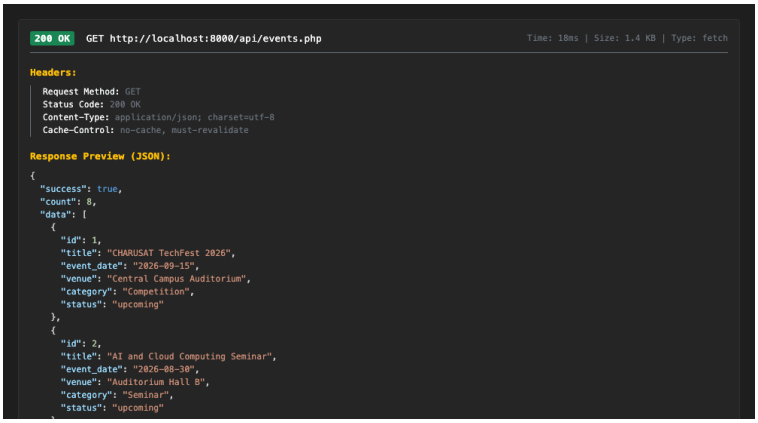


Figure 2: Browser Network Inspection Showing 200 OK Status and JSON Response Payload

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to build RESTful JSON API endpoints in PHP and consume them using the JavaScript Fetch API to deliver real-time, asynchronous CRUD operations without page reloads.